

Read-Range Extension with Semi-Passive Tags Powered by PV Cells

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Abstract

This paper presents a developed UHF-RFID tag (865–868 MHz) capable of operating in both passive and semi-passive modes. In semi-passive mode, where the tag is externally powered, the tag significantly extends its read range because of its improved sensitivity. Here, the power was provided by a photovoltaic (PV) cell, instead of the usual approach of using a battery. At an EIRP of 35 dBm, the read ranges were measured as 12.7 m and 15.0 m for vertical and horizontal polarization, respectively, representing remarkable increases of 139% and 168% compared to the passive mode. The PV cell provided enough power, even indoors at low illuminance conditions.

Ccs Concepts

• Hardware → Wireless devices • Hardware → External storage • Hardware → Reusable energy storage

Keywords

Passive tag, PV cell, RFID reader, RFID tag, Semi-passive tag, UHF-RFID

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1 Introduction

Radio Frequency Identification (RFID) systems are widely used for identification, tracking, and data collection across various industrial applications, including inventory management [1], supply chain automation [2], and health monitoring [3]. The most widely used protocol for ultra-high frequency RFID (UHF-RFID) systems is EPCglobal generation 2 (Gen 2), a standardized air interface that defines the reliable communication between RFID readers and tags. The protocol operates within the 860–960 MHz frequency band, offering long read ranges with the ability to read multiple tags simultaneously.

RFID tags can be categorized into three main types based on their power source and functional mechanism: passive, semi-passive, and active. Passive tags do not require an external power supply; instead, they harness energy from the reader signal. Semi-passive tags incorporate an external power supply for powering internal circuits, enabling larger read ranges, but still depend on the reader for communication. Active tags, which entirely rely on their external power supply, can actively transmit signals, making them suitable for applications involving long-range tracking and real-time location monitoring [4].

In [5], a UHF passive tag incorporating the UCODE 9 chip (NXP) was designed for wide-frequency applications (860–960

MHz) using a dipole patch antenna suitable for metal surfaces. In [6], a dual-band UHF passive tag with a UCODE 8 chip (NXP) was developed using a patch antenna featuring two truncated corners and an inclined I-slit. In [5] and [6], at an EIRP of 4 W (US standard), maximum read ranges of 9 m and 7.5 m were measured, respectively. In [7], a dual-operation tag using the Monza X-2K chip (Impinj) was designed that operated both as a passive and semi-passive tag. As a passive tag, a maximum read range of 3.94 m was achieved, while as a semi-passive tag, it increased to 8.82 m. In [8], a temperature sensor tag from Farsens was also tested both as a passive and semi-passive tag. As a semi-passive tag, it was supplied using a thermoelectric generator. At an EIRP of 35 dBm (EU standard), maximum read ranges of 4.4 m and 22.2 m were obtained as a passive and semi-passive tag, respectively. In [9], an active tag was designed to comply with EPCglobal Gen 2 protocols, and its read range was significantly extended to 154.6 m.

Semi-passive and active tags are usually powered using batteries; however, they have limitations, including limited energy capacity, replacement maintenance costs and environmental concerns. An alternative solution is using energy harvesting modules with a variety of energy sources such as light, thermal, and radio frequency (RF) waves [10]. Among these, light-based energy harvesting has emerged as one of the most promising approaches due to the relatively high-power density it can deliver under certain conditions. In [11], a monocrystalline silicon (c-Si) photovoltaic (PV) cell under a halogen lamp was used to power a dual-operation tag, demonstrating a significant improvement in its read range, even under low illumination levels such as 100 lux, as it achieved 21 m as semi-passive tag, while as passive tag it reached a maximum of 5.9 m. Similarly, in [12], a perovskite PV cell was used to power a semi-passive tag. The authors compared its communication range to that of a passive tag and reported a 5x increase, from less than 1 meter with the passive tag to approximately 4 meters with the PV-powered tag. These results highlight the potential of autonomous PV-RFID systems and the significant enhancement in communication range achievable with this approach.

In line with some of the previous works and as a work-in-progress, in this paper, an UHF-RFID tag was designed, implemented, and tested using a specific chip that operates in both passive (acting as a passive tag) and semi-passive (acting as a semi-passive tag) modes. In the semi-passive mode, the tag was powered using a photovoltaic (PV) cell, demonstrating a reliable operation of the tag in low illuminance conditions. The paper is organized as follows: Section 2 presents the tag design and implementation. Section 3 shows the tag power supply characterization in semi-passive mode. Section 4 theoretically calculates and experimentally characterizes the tag read range for both operation modes. The paper is concluded in Section 5.

2 Tag Design and Implementation

Fig. 1(a) shows a schematic of the tag, including a mender-shape dipole antenna integrated with UCODE G2iM+ chip from NXP.

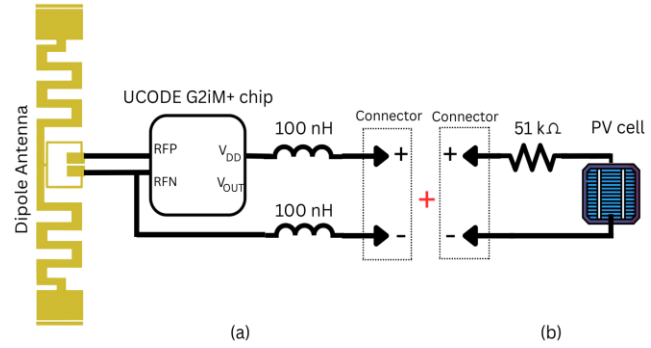


Fig. 1. The schematic of (a) the tag antenna, chip and choke inductors, and (b) the PV cell with the series resistor.

The chip operates within the UHF-RFID frequency range, uses the EPCglobal Gen 2 protocol, and has a passive and semi-passive dual-operation mode. RFP and RFN are the ungrounded and grounded antenna connectors, respectively, V_{DD} is the external power supply pin, and V_{OUT} is the output pin (unconnected here). Two choke inductors of 100 nH (B82496X001, EPCOS) were placed in series with the chip power pins (V_{DD} and RFN) to minimize RF coupling when externally powered in semi-passive mode. To power the tag in this mode, the manufacturer specifies the minimum supply current and voltage, which are at most 14 μ A and 1.8 V, respectively. An amorphous silicon Panasonic AM-1454 PV cell (Am1) with 11 cm² area was selected to provide sufficient power to the tag under indoor lighting at low illuminance (100 lux) [13]. To prevent potential damage to the RFID chip from overvoltage (> 2.2 V) and overcurrent (> 280 μ A) during high illumination levels, e.g., under direct sunlight exposure, a 51 k Ω resistor was added as shown in Fig. 1(b).

Fig. 2(a) shows a picture of the designed and implemented tag that was printed on FR4 substrate ($\tan \delta = 0.01$, $\epsilon_r = 4.6$, $\mu_r = 1$) with a thickness of 1.47 mm, and a dimension of 86.3 mm $\times$ 13.3 mm [14]. A connector was added for the external power supply. The connector, choke inductors and chip are labeled on the tag illustration. Both the PV cell and the resistor were mounted on a 60 $\times$ 50 mm² printed circuit board (PCB), as shown in Fig. 2(b), with a connector to link it to the board of Fig. 2(a).

3 Tag Power Supply Characterization

First, the experimental minimum supply current and voltage to reliable operate the tag were measured, resulting in 10 μ A and 1.55 V, respectively, which comply with the specifications. Then,

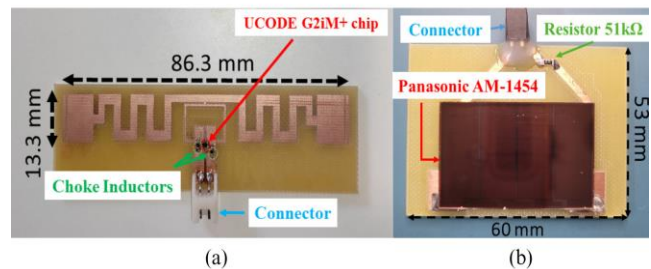


Fig. 2. Picture of (a) the implemented tag [14] and (b) the PV cell with the series resistor.

the PV cell was characterized using the system described in [13], [15]. The cell was tested at 25°C under three types of artificial light (cold and warm LED and compact fluorescent light), with illuminance ranging from 100 lux to 1500 lux, and with solar light (AM1.5) ranging from 1000 to 10000 lux. The worst-case (lowest output current and voltage from the PV cell) occurred at 100 lux with warm LED light, achieving a short circuit current of 17.7 μ A and an open circuit voltage of 2.3 V, which showcased the capabilities for powering the tag even under low illuminance conditions. Finally, the PV cell was connected to the power pins of the tag chip through the series 51 k Ω resistor, delivering 10 μ A and 1.6 V to the tag under warm LED light at 100 lux. Successful tag operation was verified.

To further assess tag operation under realistic conditions, an additional test was performed in the laboratory environment [13], which has CFL lamps at the ceiling. An illumination value of 160 lux was measured at the tag location, being the current and voltage delivered to the tag by the PV cell of 12.2 μ A at 1.6 V, respectively. Once again, successful tag operation was verified.

4 Tag Read Range Calculation and Characterization

In this section, the read range of the tag is calculated and experimentally characterized. The calculated read range (R) of a tag, derived from Friis equation, under free-space conditions, is given by [16]

$$R = \frac{\lambda}{4\pi} \sqrt{\frac{\text{EIRP} \eta_{\text{pol}}}{S_{\text{tag}}}} \quad (1)$$

where λ is the wavelength, EIRP is the effective isotropic radiated power, η_{pol} is the polarization efficiency, and S_{tag} is the tag sensitivity. For the proposed design, the 865–868 MHz band specified for UHF-RFID by the European Radiocommunications Committee (ERC) was selected. The operating frequency was set to the central value of 866.5 MHz ($\lambda = 0.346$ m) and EIRP to 35 dBm (the maximum allowed). A 3 dB polarization loss ($\eta_{\text{pol}} = 0.5$) was assumed due to the circular polarization of the reader antenna and the linear polarization of the tag antennas. The values of S_{tag} in passive and semi-passive modes were obtained from the chip datasheet and are -17.5 dBm and -27 dBm, respectively. Table 1 shows the calculated values (Calc) of R for the tag using (1) in both operation modes.

To experimentally characterize the tag read range, instead of fixing the reader's output power and increasing the distance, the

power is varied while the distance remains fixed. This procedure is commonly used in 1) real-world environments with reflections [8], [16], to avoid variations in multipath effects caused by changing distances, and 2) in anechoic chambers [17], which present minimal reflections, because of their limited size. So, we conducted outdoor measurements in “free-space” conditions, with the tag placed in the line of sight of the reader antenna at fixed distances (r_{fix}) of 1 m and 3 m in passive and semi-passive modes, respectively. A commercial reader (R420, IMPINJ) and antenna (Far-Field RAIN RFID Antenna, Laird) were used to detect the tag. The reader antenna features circular polarization, operates within a frequency range of 865–868 MHz, and has a gain (G_t) of 8.5 dBi. It was connected to the reader using a 2-meter coaxial cable (TL-092463, Laird) with a 50 Ω impedance. To manage and configure the reader, the IMPINJ ITEMTEST software was used, allowing easy control of the reader's parameters. The operating frequency of 866.3 MHz was selected from the discrete values available in the reader software, as it was the closest to 866.5 MHz. Fig. 3 showcases the measurement setup in semi-passive mode.

In each mode, the tag was positioned in both vertical (V) and horizontal (H) polarization (Pol), and, for each case, the reader output power was increased from 10 dBm in 0.25 dB steps until the tag began responding to the reader. The corresponding power level was recorded as the minimum tag activation power, denoted as $P_{t(\text{min})}$. Then, the minimum EIRP was calculated as $\text{EIRP}_{\text{min}} = P_{t(\text{min})} + G_t - L_c$, where L_c represents the cable loss (1 dB, according to the cable datasheet). Finally, the experimental read range for $\text{EIRP} = 35$ dBm, was obtained by [17]:

$$R = r_{\text{fix}} \times \sqrt{\frac{\text{EIRP}}{\text{EIRP}_{\text{min}}}} \quad (2)$$

which can be easily derived from (1). Results are shown as measured values in Table 1 (Meas). Additionally, the values of the power level received from the tag by the reader (RSSI: received signal strength indicator) are also reported.

As can be seen from the table, the measured values of R are lower than the corresponding calculated values. These differences can mainly be due to variations in our measurement setup and environmental conditions compared to those used by

Table 1. Calculations and Measurements of R in passive and semi-passive modes.

Operation Mode	Calc		Meas			
	S_{tag} (dBm)	R (m)	Pol	EIRP_{min} (dBm)	RSSI (dBm)	R (m)
Passive ($r_{\text{fix}}=1$ m)	-17.5	8.2	V	20.5	-55.5	5.3
			H	20.0	-55.5	5.6
Semi-Passive ($r_{\text{fix}}=3$ m)	-27.0	24.5	V	22.5	-69.0	12.7
			H	21.0	-68.0	15.0

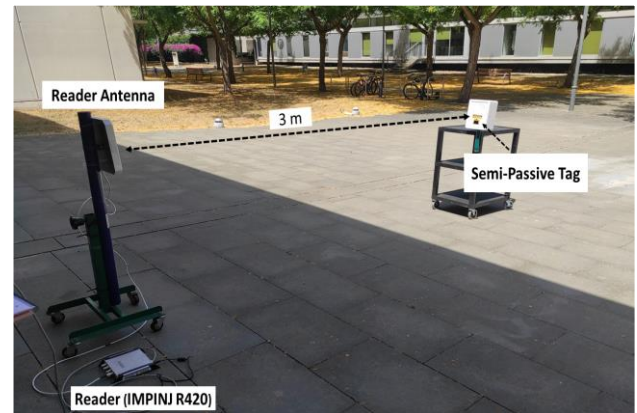


Fig. 3. Measurement setup for the tag in semi-passive mode.

manufacturers to obtain the tag sensitivity, which affects R in (1). In addition, (1) applies to an ideal free-space environment (without reflections), which is not wholly the case for the measurements. On the other hand, in both calculation and measurement results, a significant improvement is observed for R in semi-passive mode compared to passive mode: 198% for the calculation, and 139% and 168% in the measurements for V and H polarization, respectively. This is due to the highest sensitivity of the tag chip in semi-passive mode. The RSSI values in all cases are well above the reader default maximum sensitivity (-80 dBm), indicating that the read range was not limited by the reader sensitivity. The PV cell correctly powered the tag operating in semi-passive mode in all cases.

5 Conclusion

In this paper, a custom UHF-RFID tag has been presented, operating within the 865–868 MHz band and capable of functioning in both passive and semi-passive modes using the UCODE G2iM+ chip from NXP. In semi-passive mode, the read range is greatly extended because of the improved tag sensitivity. A PV cell has been used to reliably power the tag chip down to low illuminance indoor conditions of 100 lux. The measurement results have shown that, in semi-passive mode and at an EIRP of 35 dBm, the read ranges have reached 12.7 m and 15.0 m for vertical and horizontal polarizations, respectively, highlighting substantial enhancements of 139% and 168% compared to the passive mode, and confirming the effectiveness of the semi-passive design and battery-free power supply approach.

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